

NexaCore Deployment Guide

Version 3.7 | Engineering Production Delivery

Owner: Platform Engineering Effective: 01 Aug 2026 Review: 31 Jul 2027

1. Deployment Model

NexaCore uses an environment-separated delivery model: Development → Test → Staging → Production. Promotion is controlled by CI/CD policies rather than manual source copying.

1.1 Environment purpose

Environment	Purpose	Data	Deployment authority
Development	Feature development	Synthetic / masked	Developer
Test	Automated integration testing	Synthetic	CI
Staging	Production-like validation	Masked / approved test	Release owner
Production	Customer workloads	Live	Release owner + controls

1.2 Release trains

- Standard production releases are scheduled Tuesday–Thursday during approved release windows.
- Friday production releases require explicit engineering manager approval unless they are emergency/security fixes.
- Tier-1 services require a named release owner and on-call engineer.
- Production deployment windows are normally 11:00–16:00 IST for standard changes.

1.3 Change classes

Normal changes follow the standard pipeline. Emergency changes use the incident/hotfix workflow and require retrospective documentation within one business day.

2. CI/CD Pipeline and Gates

2.1 Standard pipeline

Stage	Gate	Failure action
Source checkout	Commit integrity	Stop
Build	Compilation/package success	Stop
Unit tests	Required suite pass	Stop
Static analysis	No blocking findings	Stop or exception
Security scan	Severity threshold	Stop or security exception
Integration tests	Required suites pass	Stop
Artifact publish	Checksum/version recorded	Stop
Deploy staging	Health checks	Stop
Smoke test	Critical endpoints pass	Stop
Production	Approval policy	Controlled promotion

2.2 Build reproducibility

- Build artifacts must be versioned and traceable to source commit.
- Dependency lock files should be honored in CI.
- Production artifacts should be promoted rather than rebuilt from a different source state.
- Container images must use immutable digests for production deployment.

3. Production Release Procedure

3.1 Pre-release checklist

- Confirm change ticket and owner.
- Review automated test results and deployment diff.
- Verify database migration compatibility.
- Confirm monitoring dashboards and alerts.
- Confirm rollback or forward-fix plan.
- Check on-call coverage.
- Communicate customer-impacting maintenance where required.

3.2 Deployment sequence

Step	Maximum target	Required evidence
Pre-flight	15 min	Checklist complete
Canary	10 min	Health + error metrics
Initial rollout	15 min	No regression threshold breach
Full rollout	30 min	Service healthy
Post-deploy	15 min	Smoke + dashboard review

3.3 Canary stop conditions

Pause rollout when 5xx error rate increases by more than 1 percentage point above the pre-release baseline for 5 consecutive minutes, p95 latency increases by more than 30%, or a critical health check fails.

4. Database and Infrastructure Changes

4.1 Database migration rules

- Prefer backward-compatible expand/contract migrations.
- Large backfills must be throttled and observable.
- Migration duration must be estimated before production.
- Destructive operations require explicit approval and verified backup/restore readiness.
- Application deployment should not depend on a migration completing instantaneously.

4.2 Infrastructure changes

Change	Required validation
Compute scaling	Capacity + cost estimate
Network rule	Security review + connectivity test
Storage expansion	Capacity forecast + alert threshold
IAM policy	Least privilege review
Managed service upgrade	Compatibility + rollback/restore test

4.3 Backup expectations

Tier-1 production databases target daily full backups plus transaction/log recovery according to service architecture. Restore tests are performed quarterly for Tier-1 systems and semiannually for Tier-2 systems.

5. Rollback, Incident Response and Recovery

5.1 Rollback decision

Rollback is preferred when the deployed change is the clear cause of customer impact and the previous artifact remains safe. Forward-fix is preferred when rollback would create data incompatibility or the incident is caused by an irreversible migration.

5.2 Severity response

Severity	Example	Initial response
SEV-1	Widespread customer outage / major data risk	≤ 5 min
SEV-2	Material degradation / limited outage	≤ 15 min
SEV-3	Localized issue / workaround exists	≤ 1 hr
SEV-4	Low-impact defect	Next business day

5.3 Recovery objectives

Reference targets: Tier-1 RTO 60 minutes and RPO 15 minutes; Tier-2 RTO 4 hours and RPO 1 hour. Individual services may have stricter targets documented in their service record.

5.4 Post-incident

Customer-impacting deployment incidents require a timeline, impact assessment, root/contributing factors, corrective actions and named owners. The review target is 3 business days after stabilization.

6. Deployment Metrics and Governance

6.1 FY26 release snapshot

Metric	Value
Production deployments	2,914
Successful first-attempt deployments	92.6%
Change failure rate	7.4%
Median deployment duration	18 min
Average rollback time	11 min
Tier-1 services using canary rollout	82%
Automated smoke-test coverage	91%
Deployments outside release window	3.1%

6.2 Release governance

- Weekly release-health review tracks failed deployments, rollbacks and recurring pipeline failures.
- Monthly production-readiness review checks Tier-1 service ownership and rollback evidence.
- Quarterly disaster-recovery exercise validates selected restore and failover paths.
- Deployment exceptions expire after 30 days unless renewed by the responsible engineering owner.

Document control

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